

ResearchAgent: Iterative Research Idea Generation over Scientific Literature with Large Language Models

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Abstract

The pace of scientific research, vital for improving human life, is complex, slow, and needs specialized expertise. Meanwhile, novel, impactful research often stems from both a deep understanding of prior work, and a cross-pollination of ideas across domains and fields. To enhance the productivity of researchers, we propose ResearchAgent, which leverages the encyclopedic knowledge and linguistic reasoning capabilities of Large Language Models (LLMs) to assist them in their work. This system automatically defines novel problems, proposes methods and designs experiments, while iteratively refining them based on the feedback from collaborative LLM-powered reviewing agents. Specifically, starting with a core scientific paper, ResearchAgent is augmented not only with relevant publications by connecting information over an academic graph but also entities retrieved from a knowledge store derived from shared underlying concepts mined across numerous papers. Then, mimicking a scientific approach to improving ideas with peer discussions, we leverage multiple LLM-based ReviewingAgents that provide reviews and feedback via iterative revision processes. These reviewing agents are instantiated with human preference-aligned LLMs whose criteria for evaluation are elicited from actual human judgments via LLM prompting. We experimentally validate our ResearchAgent on scientific publications across multiple disciplines, showing its effectiveness in generating novel, clear, and valid ideas based on both human and model-based evaluation results. Our initial foray into AI-mediated scientific research has important implications for the development of future systems aimed at supporting researchers in their ideation and operationalization of novel work¹.

1 Introduction

Scientific research plays a crucial role in driving innovation, advancing knowledge, solving prob-

¹Code: <https://github.com/JinheonBaek/ResearchAgent>.

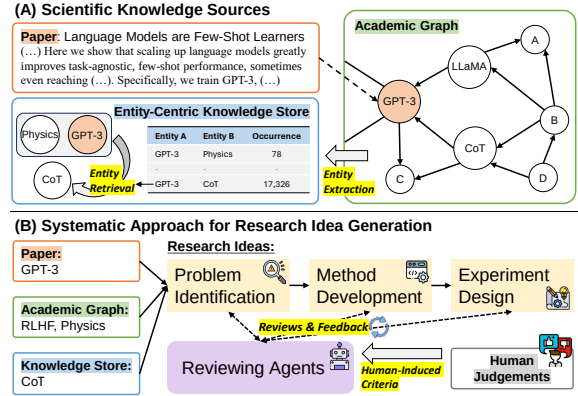


Figure 1: (A) The scientific knowledge used for research idea generation consists of a paper, its relationships over an academic graph, and entities within a knowledge store extracted from numerous papers. (B) Given them, the proposed research idea generation process involves problem identification, method development, and experiment design. Those are also iteratively refined by reviews and feedback from reviewing agents, aligned with criteria induced from human judgements.

lems, expanding our understanding of the world, and ultimately improving the lives of people in tangible ways. This process usually consists of two key components: the formulation of new research ideas and the validation of these ideas through well-crafted experiments, which are typically conducted by human researchers (Hope et al., 2023; Wang et al., 2023a; Huang et al., 2023). However, this is a slow, effort-intensive process, which requires reading and synthesizing overwhelming amounts of knowledge over the vast corpus of rapidly growing scientific literature to formulate research ideas, as well as design and perform experimental validations of those ideas. For example, the number of academic papers published per year is more than 7 million (Fire and Guestrin, 2019). Similarly, the process of testing a new pharmaceutical drug requires deep expertise, and is massively expensive and labor-intensive, often taking several years (Vamathevan et al., 2019).

In the meantime, recent Large Language Models (LLMs) (Touvron et al., 2023; OpenAI, 2023; Anil et al., 2023) have shown impressive capabili-

ties in processing and generating text with remarkable accuracy, even outperforming human experts across diverse specialized domains including math, physics, history, law, medicine, and ethics. They are able to process and analyze large volumes of data at speeds and scales far exceeding human capabilities, have internalized large swaths of human knowledge from being trained on virtually the entire web, and can identify patterns, trends, and correlations that may not be immediately apparent to human researchers (such as the usage of quantum mechanics in medical imaging or applying psychological insights in AI). This renders them ideally poised to become foundational tools to accelerate the two phases of the scientific research process: ideation of novel research opportunities, and scientific validation of those research hypotheses.

A few recent papers in the domain of LLM-augmented scientific discovery have focused on the second phase. Specifically, they attempt (Huang et al., 2023; AI4Science and Quantum, 2023; Bran et al., 2023) to mainly accelerate the experimental validation process, by writing code for machine-learning models, facilitating the exploration of chemical spaces, or advancing the simulation of molecular dynamics. Thus, in this paper, we leverage LLMs in the first phase of scientific research – specifically idea generation, whose key focus is conceptualizing novel research questions, methodologies, and experiments. To the best of our knowledge, our work is the first to leverage and evaluate the capabilities of LLMs to act as mediators in scientific idea generation in an open-ended setting.

Given our goal to build an LLM-powered ResearchAgent, we draw inspiration from how human researchers position themselves to come up with novel research ideas. We draw distinctions between three key components of their workflow: a broad and deep understanding of related scientific literature, an encyclopedic view of concepts and how they relate to one another both within and across domains, and a community of colleagues on which to rely for feedback and constructive criticism.

We model each of these three aspects in our ResearchAgent. Specifically, in order to imbibe related work, the system begins with a core scientific paper and then explores a range of related papers through references and citation relationships. Further, to develop an encyclopedic view of related concepts, we build and then augment ResearchAgent with an entity-centric knowledge store derived from co-occurrences of key concepts in the scien-

tific literature. This repository is aimed at capturing novel underlying relationships within and across domains, thereby increasing the chances of a cross-pollination of ideas (Wahle et al., 2023). Finally, to simulate robust feedback mechanisms, we instantiate a number of LLM-powered ReviewingAgents that help the ResearchAgent to iterate on research idea generation with constructive critiques. Crucially, these ReviewingAgents are prompted with evaluation criteria that are induced from real researchers’ judgements, thus aligning them with actual scientific preferential standards. An illustration of our system is provided in Figure 1.

We validate the effectiveness of ResearchAgent for research idea generation based on scientific literature across multiple disciplines. Then, on a battery of tests conducted with both human- and model-based evaluations, we demonstrate that ResearchAgent outperforms strong LLM-powered baselines by large margins, generating more clear, relevant, and significant ideas that are especially novel. Furthermore, analyses show the efficacy of our comprehensive approach to modeling ResearchAgent: the entity-centric knowledge store and the iterative idea refinement steps help the system generate meaningfully better ideas compared with an instantiation that is purely based on prior related work.

These findings highlight the immense potential of AI-mediated research assistants like ResearchAgent to enhance the ideation process in scientific research. In practice, it can support researchers by identifying knowledge gaps, proposing novel problem statements, and suggesting potential methodologies early in the research process. Also, it can assist in designing experiments and streamline the writing and refinement of research papers by generating drafts and offering feedback on how to effectively frame contributions and cite relevant work.

2 Related Work

Large Language Models LLMs have shown impressive performances across various tasks (OpenAI, 2023; Anil et al., 2023), including scientific fields such as mathematics, physics, medicine, and computer science (Portenoy et al., 2021; Romera-Paredes et al., 2023; Bran et al., 2023; Huang et al., 2023; Liu et al., 2024). For instance, GPT-4 can understand DNA sequences, design biomolecules, predict molecular behavior, and solve PDE problems (AI4Science and Quantum, 2023). However, LLMs have mainly been used for accelerating the

experimental validation of already identified research ideas, but not for identifying new problems.

Hypothesis Generation The principle of hypothesis generation is based on literature-based discovery (Swanson, 1986), which aims to discover relationships between concepts (Henry and McInnes, 2017). For instance, these concepts could be a specific disease and a compound not yet considered as a treatment for it. Early works on automatic hypothesis generation first build a corpus of discrete concepts, and then identify their relationships with machine learning approaches, e.g., using similarities between word (concept) vectors (Tshitoyan et al., 2019) or applying link prediction methods over a graph (where concepts are nodes) (Sybrandt et al., 2020; Nadkarni et al., 2021). Recent approaches are further powered by LLMs (Wang et al., 2023b; Qi et al., 2023; Yang et al., 2023), leveraging their prior knowledge about scientific disciplines. Yet, all these approaches perform idea generation in a localized manner and are designed to identify potential relationships between two variables or generate sentence-level connections, which may be sub-optimal to capture the complexity and multifaceted nature of real-world problems (e.g., urban planning involves numerous interacting variables). Meanwhile, we do not artificially restrict the target research idea to be a predictive single concept or simple binary link, instead allowing the model to generate ideas in a more open-ended fashion.

We note that there has been a recent surge of interest in exploring scientific idea generation: from Li et al. (2024) that focus on evaluating whether LLMs can generate research ideas that are better than human ideas, to Lu et al. (2024) that aim to automatically generate full research papers (including idea development, code writing, and experiment execution), to Li et al. (2024) that enhance the idea generation process by organizing a sequential chain of literature, all of which acknowledge and build upon insights from a prior version of our paper.

Knowledge-Augmented LLMs The approach to augment LLMs with external knowledge makes them more accurate and relevant to target contexts. Much prior work aims at improving the factuality of LLM responses to queries by retrieving the relevant documents and injecting them into the LLM input (Lazaridou et al., 2022; Ram et al., 2023; Shi et al., 2023). In addition, given that entities or facts are atomic units for representing knowledge, recent studies augment LLMs with them (Baek et al.,

2023; Wu et al., 2023). In contrast to these efforts, which use knowledge units piecemeal, we instead jointly leverage accumulated knowledge over massive troves of scientific papers. Also, Baek et al. (2024) proposes to use entities for query suggestion, which – while similar – has the entirely different objective of narrowing the focus of LLMs to entities already present in their context. Instead, our approach retrieves and integrates entities outside the given context, enabling LLMs to explore other concepts for research idea generation.

Iterative Refinements with LLMs Similar to humans, LLMs do not always generate optimal outputs on their first attempt. To tackle this, drawing inspiration from humans who can iteratively refine their thoughts based on critiques from themselves and their peers, many recent studies have investigated the potential of LLMs to correct and refine their outputs, demonstrating that they indeed possess those capabilities (Welleck et al., 2023; Madaan et al., 2023; Shridhar et al., 2023; Ganguli et al., 2023; Wang et al., 2023b; Qi et al., 2023; Yang et al., 2023). Based on their findings, we extend this paradigm (and further test their capability) to our novel scenario of research idea generation.

3 Method

We present ResearchAgent, a system that automatically proposes research ideas with LLMs.

3.1 LLM-Powered Research Idea Generation

We begin by formally introducing the new problem of research idea generation, followed by an explanation of how LLMs are utilized to tackle it.

Research Idea Generation The goal of the research idea generation task is to formulate new and valid research ideas, to enhance the overall efficiency of the first phase of scientific discovery. While we acknowledge that the real process by which humans conduct research is varied and complex to an extent well beyond the scope of this scientific study, we attempt to model simulacra in three systematic steps that would likely be maximally beneficial to a researcher seeking assistance from an AI system. These are namely, identifying novel research ideas, proposing methods to validate these ideas, and designing experiments to measure the success of these methods in relation to the ideas.

To accomplish the aforementioned steps, we utilize the existing literature (such as academic pub-